4-7 THREE-PHASE HALF-WAVE CONVERTERS

Three-phase converters provide higher average output voltage, and in addition the frequency of the ripples on the output voltage is higher compared to that of single-phase converters. As a result, the filtering requirements for smoothing out the load current is simpler. For these reasons, three-phase converters are used extensively in high-power variable-speed drives. Three single-phase half-wave converters in Fig. 4-1a can be connected to form a three-phase half-wave converter, as shown in Fig. 4-7a.

When thyristor T_1 is fired at $\omega t = \pi/6 + \alpha$, the phase voltage v_{an} appears across the load until thyristor T_2 is fired at $\omega t = 5\pi/6 + \alpha$. When thyristor T_2 is fired, thyristor T_1 is reverse biased, because the line-to-line voltage, v_{ab} ($= v_{an} - v_{bn}$), is negative and T_1 is turned off. The phase voltage v_{bn} appears across the load until thyristor T_3 is fired at $\omega t = 3\pi/2 + \alpha$. When thyristor T_3 is fired, T_2 is turned off and v_{cn} appears across the load until T_1 is fired again at the beginning of next cycle. Figure 4-7b shows the $v - i$ characteristics of the load and this is a one-quadrant converter. Figure 4-7c shows the input voltages, output voltage, and the current through thyristor T_1 for a highly inductive load. For a resistive load and $\alpha > \pi/6$, the load current would be discontinuous and each thyristor is self-commutated when the polarity of its phase voltage is reversed. The frequency of output ripple voltage is $3f_s$. This converter is not normally used in practical systems, because the supply currents contain dc components.

If the phase voltage is $v_{an} = V_m \sin \omega t$, the average output voltage for a continuous load current is

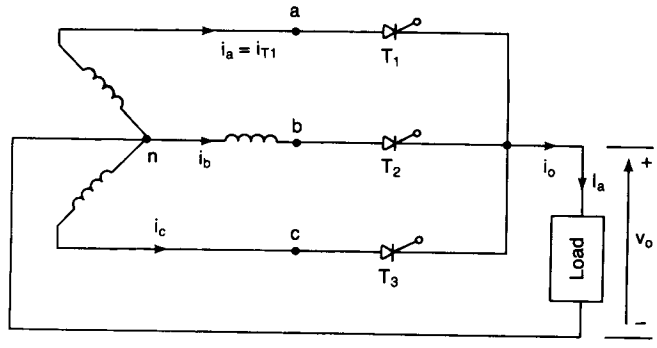
$$V_{dc} = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{5\pi/6+\alpha} V_m \sin \omega t \, d(\omega t) = \frac{3\sqrt{3}\, V_m}{2\pi} \cos \alpha \qquad (4\text{-}51)$$

where V_m is the peak phase voltage. The maximum average output voltage that occurs at delay angle, $\alpha = 0$ is

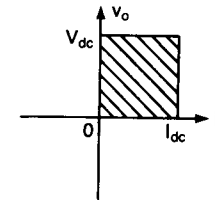
$$V_{dm} = \frac{3\sqrt{3}\, V_m}{2\pi}$$

and the normalized average output voltage is

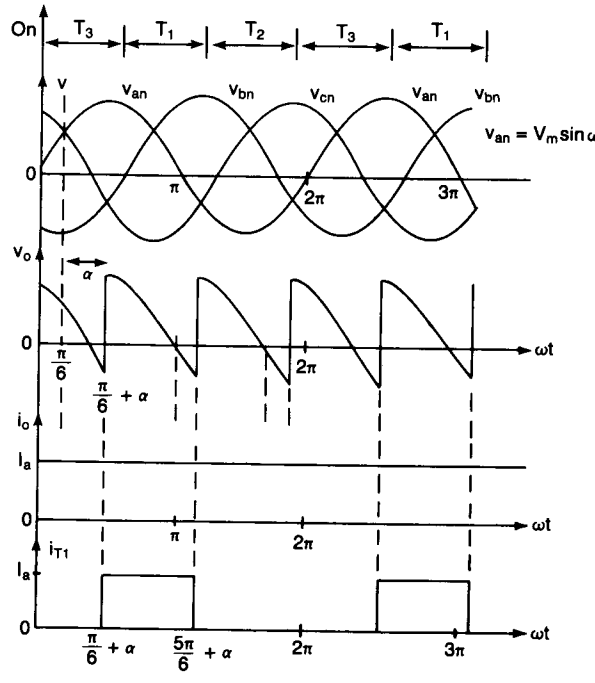
$$V_n = \frac{V_{dc}}{V_{dm}} = \cos \alpha \qquad (4\text{-}52)$$



(a) Circuit



(b) Quadrant



(c) For inductive load

Figure 4-7 Three-phase half-wave converter.

The rms output voltage is found from

$$V_{\text{rms}} = \left[\frac{3}{2\pi}\left[\int_{\pi/6+\alpha}^{5\pi/6+\alpha} V_m^2 \sin^2 \omega t \, d(\omega t)\right]\right]^{1/2}$$

$$= \sqrt{3}\, V_m \left(\frac{1}{6} + \frac{\sqrt{3}}{8\pi}\cos 2\alpha\right)^{1/2} \tag{4-53}$$

For a resistive load and $\alpha \geq \pi/6$:

$$V_{dc} = \frac{3}{2\pi}\int_{\pi/6+\alpha}^{\pi} V_m \sin\omega t\, d(\omega t) = \frac{3V_m}{2\pi}\left[1 + \cos\left(\frac{\pi}{6} + \alpha\right)\right] \tag{4-51a}$$

$$V_n = \frac{V_{dc}}{V_{dm}} = \frac{1}{\sqrt{3}}\left[1 + \cos\left(\frac{\pi}{6} + \alpha\right)\right] \tag{4-52a}$$

$$V_{rms} = \frac{3}{2\pi}\left[\int_{\pi/6+\alpha}^{\pi} V_m^2 \sin^2\omega t\, d(\omega t)\right]^{1/2}$$

$$= \sqrt{3}V_m\left[\frac{5}{24} - \frac{\alpha}{4\pi} + \frac{1}{8\pi}\sin\left(\frac{\pi}{3} + 2\alpha\right)\right]^{1/2} \tag{4-53a}$$

Example 4-6

A three-phase half-wave converter in Fig. 4-7a is operated from a three-phase Y-connected 208-V 60-Hz supply and the load resistance is $R = 10\ \Omega$. If it is required to obtain an average output voltage of 50% of the maximum possible output voltage, calculate the (a) delay angle α; (b) rms and average output currents; (c) average and rms thyristor currents; (d) rectification efficiency; (e) transformer utilization factor, TUF; and (f) input power factor, PF.

Solution The phase voltage is $V_s = 208/\sqrt{3} = 120.1$ V, $V_m = \sqrt{2}\,V_s = 169.83$ V, $V_n = 0.5$, and $R = 10\ \Omega$. The maximum output voltage is

$$V_{dm} = \frac{3\sqrt{3}\,V_m}{2\pi} = 3\sqrt{3} \times \frac{169.83}{2\pi} = 140.45 \text{ V}$$

The average output voltage, $V_{dc} = 0.5 \times 140.45 = 70.23$ V.

(a) For a resistive load, the load current is continuous if $\alpha \leq \pi/6$ and Eq. (4-52) gives $V_n \geq \cos(\pi/6) = 86.6\%$. With a resistive load and 50% output, the load current is discontinuous. From Eq. (4-52a), $0.5 = (1/\sqrt{3})[1 + \cos(\pi/6 + \alpha)]$ and the delay angle is $\alpha = 67.7°$.

(b) The average output current, $I_{dc} = V_{dc}/R = 70.23/10 = 7.02$ A. From Eq. (4-53a), $V_{rms} = 94.74$ V and the rms load current, $I_{rms} = 94.74/10 = 9.47$ A.

(c) The average current of a thyristor, $I_{DT} = I_{dc}/3 = 7.02/3 = 2.34$ A and the rms current of a thyristor, $I_{RT} = I_{rms}/\sqrt{3} = 9.47/\sqrt{3} = 5.47$ A.

(d) From Eq. (2-44) the rectification efficiency is $= 70.23 \times 7.02/(94.74 \times 9.47) = 54.95\%$.

(e) The rms input line current is the same as the thyristor rms current, and the input volt-ampere rating, $VI = 3V_sI_s = 3 \times 120.1 \times 5.47 = 1970.84$ W. From Eq. (2-49), TUF $= 70.23 \times 7.02/1970.84 = 0.25$ or 25%.

(f) The output power, $P_o = I_{rms}^2 R = 9.47^2 \times 10 = 896.81$ W. The input power factor, PF $= 896.81/1970.84 = 0.455$ (lagging).

Note. Due to the delay angle, α, the fundamental component of input line current is also delayed with respect to the input phase voltage.

2-9 MULTIPHASE STAR RECTIFIERS

We have seen in Eq. (2-59) that the average output voltage which could be obtained from single-phase full-wave rectifiers is $0.6366V_m$ and these rectifiers are used in applications up to the 15-kW power level. For larger power output, *three-phase* and *multiphase* rectifiers are used. The Fourier series of the output voltage given by Eq. (2-63) indicates that the output contains harmonics and the frequency of the *fundamental component* is two times the source frequency ($2f$). In practice, a filter is normally used to reduce the level of harmonics in the load; and the size of the filter decreases with the frequency of the harmonics. In addition to the larger power output of multiphase rectifiers, the fundamental frequency of the harmonics is also increased and is q times the source frequency (qf).

The rectifier circuit of Fig. 2-14(a) can be extended to multiple phases by having multiphase windings on the transformer secondary as shown in Fig. 2-17a. This circuit may be considered as q single-phase half-wave rectifiers and can be considered as a half-wave type. The kth diode will conduct during the period when the voltage of kth phase is higher than that of other phases. The waveforms for the voltages and currents are shown in Fig. 2-17b. The conduction period of each diode is $2\pi/q$.

It can be noticed from Fig. 2-17b that the current flowing through the secondary winding is unidirectional and contains a dc component. Only one secondary winding carries current at a particular time, and as a result the primary must be connected in delta in order to eliminate the dc component in the input side of the transformer.

The average output voltage for a q-phase rectifier is given by

$$V_{dc} = \frac{2}{2\pi/q}\int_0^{\pi/q} V_m \cos \omega t \, d(\omega t) = V_m \frac{q}{\pi}\sin\frac{\pi}{q} \tag{2-69}$$

$$V_{rms} = \left[\frac{2}{2\pi/q}\int_0^{\pi/q} V_m^2 \cos^2 \omega t \, d\,(\omega t)\right]^{1/2}$$

$$= V_m\left[\frac{q}{2\pi}\left(\frac{\pi}{q} + \frac{1}{2}\sin\frac{2\pi}{q}\right)\right]^{1/2} \tag{2-70}$$

If the load is purely resistive, the peak current through a diode is $I_m = V_m/R$ and we can find the rms value of a diode current (or transformer secondary current)

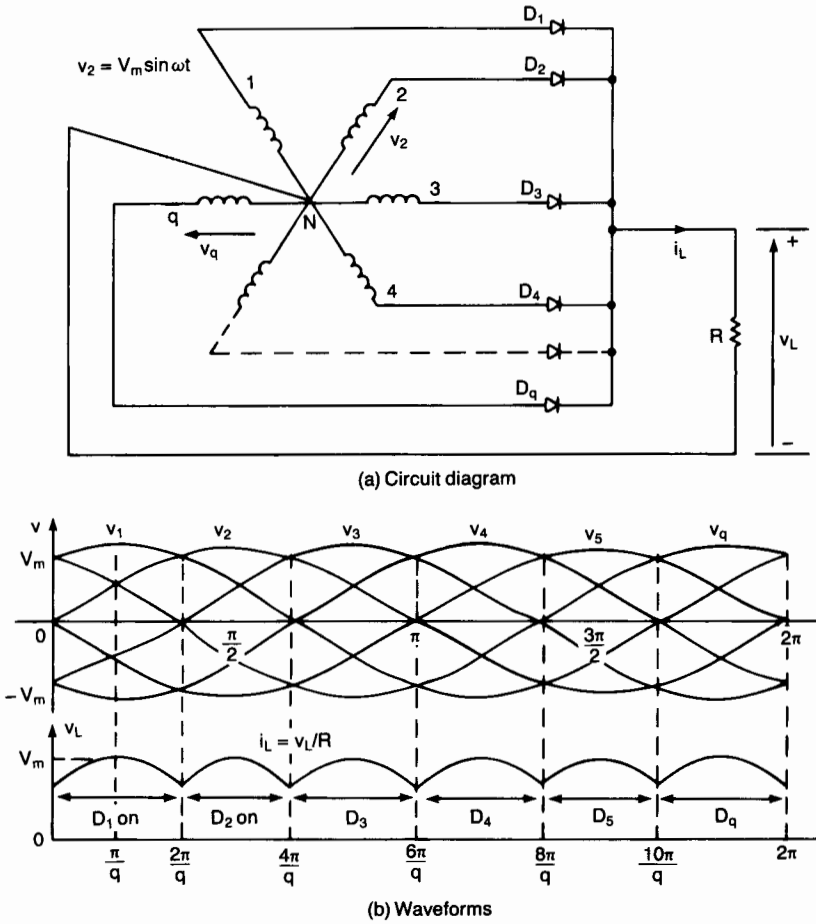


Figure 2-17 Multiphase rectifiers.

as

$$I_s = \left[\frac{2}{2\pi}\int_0^{\pi/q} I_m^2 \cos^2 \omega t \, d\,(\omega t)\right]^{1/2}$$

$$= I_m \left[\frac{1}{2\pi}\left(\frac{\pi}{q} + \frac{1}{2}\sin\frac{2\pi}{q}\right)\right]^{1/2} \tag{2-71}$$

Example 2-10

A three-phase star rectifier has a purely resistive load with R ohms. Determine the (a) efficiency; (b) form factor; (c) ripple factor; (d) transformer utilization factor; (e) peak inverse voltage (PIV) of each diode; and (f) peak current through a diode if the rectifier delivers $I_{dc} = 30$ A at a output voltage of $V_{dc} = 140$ V.

Solution For a three-phase rectifier $q = 3$ in Eqs. (2-69), (2-70), and (2-71).

(a) From Eq. (2-69), $V_{dc} = 0.827V_m$ and $I_{dc} = 0.827V_m/R$. From Eq. (2-70), $V_{rms} = 0.84068V_m$ and $I_{rms} = 0.84068V_m/R$. From Eq. (2-42), $P_{dc} = (0.827V_m)^2/R$, from Eq. (2-43), $P_{ac} = (0.84068V_m)^2/R$, and from Eq. (2-44), the efficiency,

$$\eta = \frac{(0.827V_m)^2}{(0.84068V_m)^2} = 96.77\%$$

(b) From Eq. (2-46), the form factor, FF = 0.84068/0.827 = 1.0165 or 101.65%.

(c) From Eq. (2-48), the ripple factor, RF = $\sqrt{1.0165^2 - 1} = 0.1824 =$ 18.24%.

(d) From Eq. (2-57), the rms voltage of the transformer secondary, $V_s = 0.707V_m$. From Eq. (2-71), the rms current of the transformer secondary,

$$I_s = 0.4854I_m = \frac{0.4854V_m}{R}$$

The volt-ampere rating (VA) of the transformer is

$$\text{VA} = 3V_sI_s = 3 \times 0.707V_m \times \frac{0.4854V_m}{R}$$

From Eq. (2-49),

$$\text{TUF} = \frac{0.827^2}{3 \times 0.707 \times 0.4854} = 0.6643$$

(e) The peak inverse voltage of each diode is equal to the peak value of the secondary line-to-line voltage. Three-phase circuits are reviewed in Appendix A. The line-to-line voltage is $\sqrt{3}$ times the phase voltage and thus PIV = $\sqrt{3}V_m$.

(f) The average current through each diode is

$$I_d = \frac{2}{2\pi}\int_0^{\pi/q} I_m \cos \omega t \, d(\omega t) = I_m \frac{1}{\pi} \sin \frac{\pi}{q} \tag{2-72}$$

For $q = 3$, $I_d = 0.2757I_m$. The average current through each diode is $I_d = 30/3 =$ 10 A and this gives the peak current as $I_m = 10/0.2757 = 36.27$ A.

Example 2-11

(a) Express the output voltage of a q-phase rectifier in Fig. 2-17a in Fourier series. (b) If $q = 6$, $V_m = 170$ V and the supply frequency is $f = 60$ Hz, determine the rms value of the dominant harmonic and its frequency.

Solution (a) The waveforms for q-pulses are shown in Fig. 2-17b and the frequency of the output is q times the fundamental component (qf). To find the constants of the Fourier series, we integrate from $-\pi/q$ to π/q and the constants are

$$b_n = 0$$

$$a_n = \frac{1}{\pi/q}\int_{-\pi/q}^{\pi/q} V_m \cos \omega t \cos n\omega t \, d(\omega t)$$

$$= \frac{qV_m}{\pi}\left\{\frac{\sin [(n-1)\pi/q]}{n-1} + \frac{\sin [(n+1)\pi/q]}{n+1}\right\}$$

$$= \frac{qV_m}{\pi}\frac{(n+1)\sin [(n-1)\pi/q] + (n-1)\sin [(n+1)\pi/q]}{n^2-1}$$

After simplification and then using trigonometric relationships

$$\sin (A + B) = \sin A \cos B + \cos A \sin B$$

and

$$\sin (A - B) = \sin A \cos B - \cos A \sin B$$

we get

$$a_n = \frac{2qV_m}{\pi(n^2 - 1)} \left(n \sin \frac{n\pi}{q} \cos \frac{\pi}{q} - \cos \frac{n\pi}{q} \sin \frac{\pi}{q} \right) \tag{2-73}$$

For a rectifier with q pulses per cycle, the harmonics of the output voltage are: qth, $2q$th, $3q$th, $4q$th and Eq. (2-73) is valid for $n = 0, 1q, 2q, 3q$. The term $\sin (n\pi/q) = \sin \pi = 0$ and Eq. (2-73) becomes

$$a_n = \frac{-2qV_m}{\pi(n^2 - 1)} \left(\cos \frac{n\pi}{q} \sin \frac{\pi}{q} \right)$$

The dc component is found by letting $n = 0$ and is

$$V_{dc} = \frac{a_0}{2} = V_m \frac{q}{\pi} \sin \frac{\pi}{q} \tag{2-74}$$

which is the same as Eq. (2-69). The Fourier series of the output voltage is expressed as

$$v_L(t) = \frac{a_0}{2} + \sum_{n=q,2q,\ldots}^{\infty} a_n \cos n\omega t$$

Substituting the value of a_n, we obtain

$$v_L = V_m \frac{q}{\pi} \sin \frac{\pi}{q} \left(1 - \sum_{n=q,2q,\ldots}^{\infty} \frac{2}{n^2 - 1} \cos \frac{n\pi}{q} \cos n\omega t \right) \tag{2-75}$$

(b) For $q = 6$, the output voltage is expressed as

$$v_L(t) = 0.9549V_m \left(1 + \frac{2}{35} \cos 6\omega t - \frac{2}{143} \cos 12\omega t + \cdots \right) \tag{2-76}$$

The sixth harmonic is the dominant one. The rms value of a sinusoidal voltage is $1/\sqrt{2}$ times its peak magnitude, and the rms of the sixth harmonic is $V_6 = V_m \times 2 \times 0.9549/(35 \times \sqrt{2}) = 6.56$ A and its frequency is $f_6 = 6f = 360$ Hz.